



UNIT - 3

Cloud Computing

Cloud Computing

Cloud Computing Architecture and Management

- ✓ Cloud architecture
- ✓ Layer, Anatomy of the Cloud
- ✓ Network Connectivity in Cloud Computing
- ✓ Applications on the Cloud
- ✓ Managing the Cloud
- ✓ Managing the Cloud Infrastructure
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Introduction

There are several processes and components of cloud computing that need to be discussed.

One of the topics of such prime importance is architecture.

Architecture is the hierarchical view of describing a technology. This usually includes the components over which the existing technology is built and the components that are dependent on the technology.

Another topic that is related to architecture is anatomy.

Anatomy describes the core structure of the cloud. Once the structure of the cloud is clear, the network connections in the cloud and the details about the cloud application need to be known. This is important as the cloud is a completely Internet-dependent technology..

Continue...

Similarly, cloud management discusses the important management issues and ways in which the current cloud scenario is managed. It describes the way an application and infrastructure in the cloud are managed.

Management is important because of the quality of service (QoS) factors that are involved in the cloud.

These QoS factors form the basis for cloud computing. All the services are given based on these QoS factors.

Similarly, application migration to the cloud also plays a very important role. Not all applications can be directly deployed to the cloud.

An application needs to be properly migrated to the cloud to be considered a proper cloud application that will have all the properties of the cloud.

Cloud Architecture

Any technological model consists of an architecture based on which the model functions, which is a hierarchical view of describing the technology.

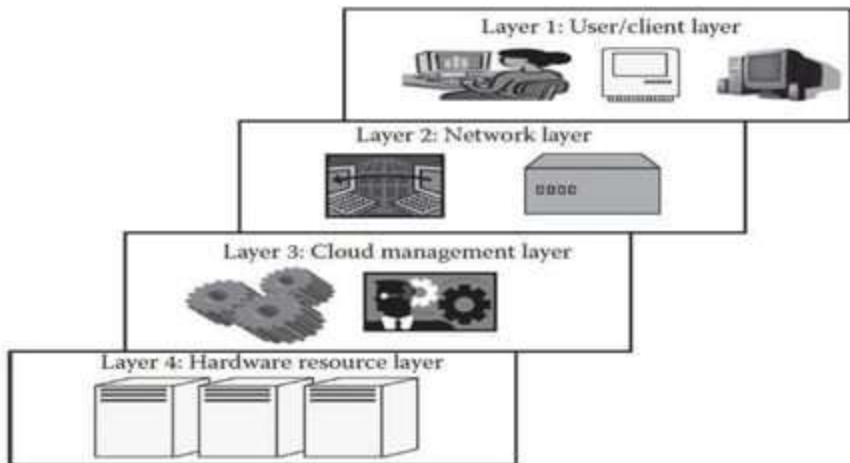
The cloud also has an architecture that describes its working mechanism.

It includes the dependencies on which it works and the components that work over it.

The cloud is a recent technology that is completely dependent on the Internet for its functioning.

The cloud architecture can be divided into four layers based on the access of the cloud by the user. They are as follows.

Cloud Architecture



Layer 1 (User/Client Layer)

This layer is the lowest layer in the cloud architecture.

All the users or client belong to this layer.

This is the place where the client/user initiates the connection to the cloud.

The client can be any device such as a thin client, thick client, or mobile or any handheld device that would support basic functionalities to access a web application.

The thin client here refers to a device that is completely dependent on some other system for its complete functionality. In simple terms, they have very low processing capability..

Layer 1 (User/Client Layer)

Similarly, thick clients are general computers that have adequate processing capability.

They have sufficient capability for independent work.

Usually, a cloud application can be accessed in the same way as a web application.

But internally, the properties of cloud applications are significantly different.

Layer 2 (Network Layer)

This layer allows the users to connect to the cloud.

The whole cloud infrastructure is dependent on this connection where the services are offered to the customers.

This is primarily the Internet in the case of a public cloud.

The public cloud usually exists in a specific location and the user would not know the location as it is abstract.

public cloud can be accessed all over the world.

In the case of a private cloud, the connectivity may be provided by a local area network (LAN). Even in this case, the cloud completely depends on the network that is used.

Layer 2 (Network Layer)

Usually, when accessing the public or private cloud, the users require minimum bandwidth, which is sometimes defined by the cloud providers.

This layer does not come under the purview of service-level agreements (SLAs), that is, SLAs do not take into account the Internet connection between the user and cloud for quality of service (QoS).

Layer 3 (Cloud Management Layer)

This layer consists of softwares that are used in managing the cloud.

The softwares can be a cloud operating system (OS), a software that acts as an interface between the data center (actual resources) and the user, or a management software that allows managing resources.

These softwares usually allow resource management (scheduling, provisioning, etc.), optimization (server consolidation, storage workload consolidation), and internal cloud governance.

This layer comes under the purview of SLAs, that is, the operations taking place in this layer would affect the SLAs that are being decided upon between the users and the service providers.

Layer 3 (Cloud Management Layer)

Any delay in processing or any discrepancy in service provisioning may lead to an SLA violation.

As per rules, any SLA violation would result in a penalty to be given by the service provider.

These SLAs are for both private and public clouds Popular service providers are Amazon Web Services (AWS) and Microsoft Azure for public cloud. Similarly, OpenStack and Eucalyptus allow private cloud creation, deployment, and management.

Layer 4 (Hardware Resource Layer)

Layer 4 consists of provisions for actual hardware resources.

Usually, in the case of a public cloud, a data center is used in the back end.

Similarly, in a private cloud, it can be a data center, which is a huge collection of hardware resources interconnected to each other that is present in a specific location or a high configuration system.

This layer comes under the purview of SLAs. This is the most important layer that governs the SLAs.

Layer 4 (Hardware Resource Layer)

Whenever a user accesses the cloud, it should be available to the users as quickly as possible and should be within the time that is defined by the SLAs.

As mentioned, if there is any discrepancy in provisioning the resources or application, the service provider has to pay the penalty.

Hence, the data center consists of a high-speed network connection and a highly efficient algorithm to transfer the data from the data center to the manager.

There can be a number of data centers for a cloud, and similarly, a number of clouds can share a data center.

Layer 4 (Hardware Resource Layer)

The architecture of a cloud layering is strict, and for any cloud application, this is followed.

There can be a little loose isolation between layer 3 and layer 4 depending on the way the cloud is deployed.

Anatomy of the Cloud

Cloud anatomy can be simply defined as the structure of the cloud.

Cloud anatomy cannot be considered the same as cloud architecture.

It may not include any dependency on which or over which the technology works,

whereas architecture wholly defines and describes the technology over which it is working.

Architecture is a hierarchical structural view that defines the technology as well as the technology over which it is dependent or/and the technology that are dependent on it.

Anatomy of the Cloud

Anatomy can be considered as a part of architecture.

The basic structure of the cloud is described, elaborated, and minute structural details can be given.

Below Figure depicts the most standard anatomy that is the base for the cloud.

It depends on the person to choose the depth of description of the cloud.

Anatomy of the Cloud

There are basically five components of the cloud

Application
Platform
Virtualized infrastructure
Virtualization
Server/storage/datacenters

Cloud Structure

Anatomy of the Cloud

1. **Application:** *The upper layer is the application layer. In this layer, any applications are executed.*
2. **Platform:** *This component consists of platforms that are responsible for the execution of the application. This platform is between the infrastructure and the application.*
3. **Infrastructure:** *The infrastructure consists of resources over which the other components work. This provides computational capability to the user.*

Anatomy of the Cloud

4. Virtualization: *Virtualization is the process of making logical components of resources over the existing physical resources. The logical components are isolated and independent, which form the infrastructure*

5. Physical hardware: *The physical hardware is provided by server and storage units.*

Network Connectivity in Cloud Computing

Cloud computing is a technique of resource sharing where servers, storage, and other computing infrastructure in multiple locations are connected by networks.

In the cloud, when an application is submitted for its execution, needy and suitable resources are allocated from this collection of resources.

These resources are connected via the Internet, the users get their required results.

For many cloud computing applications, network performance will be the key issue to cloud computing performance.

Since cloud computing has various deployment options, we now consider the important aspects related to the cloud deployment models and their accessibility from the viewpoint of network connectivity.

Public Cloud Access Networking

In this option, the connectivity is often through the Internet, though some cloud providers may be able to support virtual private networks (VPNs) for customers.

Accessing public cloud services will always create issues related to security, which in turn is related to performance.

One of the possible approaches toward the support of security is to promote connectivity through encrypted tunnels, so that the information may be sent via secure pipes on the Internet.

This procedure will be an overhead in the connectivity, and using it will certainly increase delay and may impact performance.

Public Cloud Access Networking

If we want to reduce the delay without compromising security, then we have to select a suitable routing method.

such as the one reducing the delay by minimizing transit *hops in the end-to-end connectivity between the cloud provider and cloud consumer.*

Since the end-to-end connectivity support is via the Internet, which is a complex federation of interconnected providers (known as Internet service providers [ISPs]), one has to look at the options of selecting the path.

Private Cloud Access Networking

In the private cloud deployment model, since the cloud is part of an organizational network, the technology and approaches are local to the in-house network structure.

This may include an Internet VPN or VPN service from a network operator.

If the application access was properly done with an organizational network—connectivity in a *precloud configuration*—*transition to private cloud computing will not affect the access performance.*

Intracloud Networking for Public Cloud Services

Another network connectivity consideration in cloud computing is intracloud networking for public cloud services.

Here, the resources of the cloud provider and thus the cloud service to the customer are based on the resources that are geographically apart from each other but still connected via the Internet.

Public cloud computing networks are internal to the service provider and thus not visible to the user/customer;

however, the security aspects of connectivity and the access mechanisms of the resources are important. Another issue to look for is the QoS in the connected resources worldwide.

Most of the performance issues and violations from these are addressed in the SLAs commercially.

Private Intracloud Networking

The most complicated issue for networking and connectivity in cloud computing is private intracloud networking.

What makes this particular issue so complex is that it depends on how much intracloud connectivity is associated with the applications being executed in this environment.

Private intracloud networking is usually supported over connectivity between the major data center sites owned by the company.

At a minimum, all cloud computing implementations will rely on intracloud networking to link users with the resource to which their application was assigned.

Private Intracloud Networking

Once the resource linkage is made, the extent to which intracloud networking is used depends on whether the application is componentized based on *service-oriented architecture (SOA)* or not, *among multiple systems*.

If the principle of SOA is followed, then traffic may move between components of the application, as well as between the application and the user.

The performance of those connections will then impact cloud computing performance overall.

the impact of cloud computing performance is the differences that exist between the current application and the network relationships with the application.

Private Intracloud Networking

There are reasons to consider the networks and connectivity in cloud computing with newer approaches as globalization and changing network requirements, especially those related to increased Internet usage, are demanding more flexibility in the network architectures of today's enterprises.

New Facets in Private Networks

Conventional private networks have been architected for on-premise applications and maximum Internet security. Typically, applications such as e-mail, file sharing, and *enterprise resource planning (ERP) systems* are delivered to on-premise-based servers at each corporate data center.

Increasingly today, software vendors are offering Software as a Service (SaaS) as an alternative for their software support to the corporate offices, which brings more challenges in the access and usage mechanisms of software from data center servers and in the connectivity of network architectures.

New Facets in Private Networks

The traditional network architecture for these global enterprises was not designed to optimize performance for cloud applications, now that many applications including mission-critical applications are transitioning (moving) from on-premise based to cloud based.

Path for Internet Traffic

The traditional Internet traffic through a limited set of Internet gateways poses performance and availability issues for end users who are using cloud-based applications.

It can be improved if a more widely distributed Internet gateway infrastructure and connectivity are being supported for accessing applications, as they will provide lower-latency access to their cloud applications.

As the volume of traffic to cloud applications grows, the percentage of the legacy network's capacity in terms of traffic to regional gateways increases.

Applications such as video conferencing would hog more bandwidth while mission-critical applications such as ERP will consume less bandwidth, and hence, one has to plan a correct connectivity and path between providers and consumers.

Applications on the Cloud

The power of a computer is realized through the applications.

There are several types of applications. The first type of applications that was developed and used was a stand-alone application.

A stand-alone application is developed to be run on a single system that does not use network for its functioning.

These stand-alone systems use only the machine in which they are installed.

The functioning of these kinds of systems is totally dependent on the resources or features available within the system.

These systems do not need the data or processing power of other systems; they are self-sustaining.

Applications on the Cloud

But as the time passed, the requirements of the users changed and certain applications were required, which could be accessed by other users away from the systems. This led to the inception of web application.

The web applications were different from the stand-alone applications in many aspects.

The main difference was the client server architecture that was followed by the web application.

Unlike stand-alone applications, these systems were totally dependent on the network for its working.

Here, there are basically two components, called as the client and the server.

Applications on the Cloud

The server is a high-end machine that consists of the web application installed.

This web application is accessed from other client systems.

The client can reside anywhere in the network.

It can access the web application through the Internet.

This type of application was very useful, and this is extensively used from its inception and now has become an important part of day-to-day life.

Though this application is much used, there are shortcomings as discussed in the following:

Shortcomings of Web Application

- ✓ The web application is not elastic and cannot handle very heavy loads, that is, it cannot serve highly varying loads.
- ✓ The web application is not multitenant.
- ✓ The web application does not provide a quantitative measurement of the services that are given to the users, though they can monitor the user.
- ✓ The web applications are usually in one particular platform.
- ✓ The web applications are not provided on a pay-as-you-go basis; thus, a particular service is given to the user for permanent or trial use and usually the timings of user access cannot be monitored.
- ✓ Due to its nonelastic nature, peak load transactions cannot be handled.

Computer Application evolution

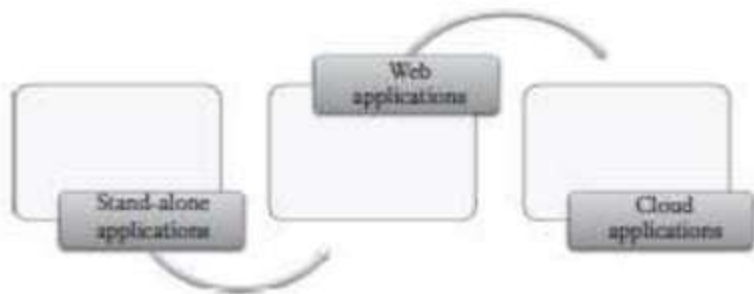


FIGURE 3.3
Computer application evolution.

Cloud Application

Primarily to solve the previously mentioned problem, the cloud applications were developed. Figure 3.3 depicts the improvements in the applications.

The cloud as mentioned can be classified into three broad access or service models, Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS).

Cloud application in general refers to a SaaS application.

A cloud application is different from other applications; they have unique features.

A cloud application usually can be accessed as a web application but its properties differ.

Cloud Application

According to NIST [3], the features that make cloud applications unique are described in the following (Figure 3.4 depicts the features of a cloud application):

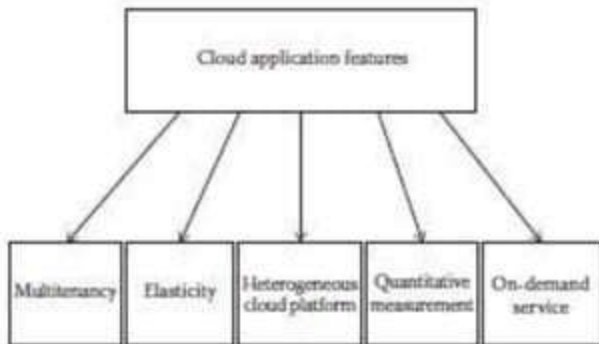


FIGURE 3.4
Features of cloud.

Cloud Application

1. **Multitenancy:** *Multitenancy is one of the important properties of cloud that make it different from other types of application in which the software can be shared by different users with full independence.*

Here, independence refers to logical independence.

Each user will have a separate application instance and the changes in one application would not affect the other.

Physically, the software is shared and is not independent. The degree of physical isolation is very less.

The logical independence is what is guaranteed. There are no restrictions in the number of applications being shared.

Cloud Application

The difficulty in providing logical isolation depends on the physical isolation to a certain extent.

If an application is physically too close, then it becomes difficult to provide multitenancy.

Web application and cloud application are similar as the users use the same way to access both.

Figure 3.5 depicts a multitenant application where several users share the same application.

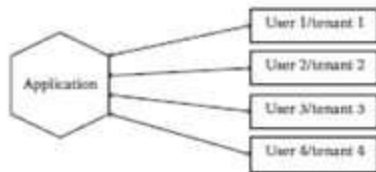


FIGURE 3.5
Multitenancy

Cloud Application

2. **Elasticity:** Elasticity is also a unique property that enables the cloud to serve better.

elasticity can be defined as the degree to which a system is able to adapt to workload changes by provisioning and deprovisioning resources in an autonomic manner such that at each point in time, the available resources match the current demand as closely as possible.

Elasticity allows the cloud providers to efficiently handle the number of users, from one to several hundreds of users at a time.

it supports the rapid fluctuation of loads, that is, the increase or decrease in the number of users and their usage can rapidly change.

Cloud Application

3. Heterogeneous cloud platform:

The cloud platform supports heterogeneity, wherein any type of application can be deployed in the cloud.

Because of this property, the cloud is flexible for the developers, which facilitates deployment.

The applications that are usually deployed can be accessed by the users using a web browser.

Cloud Application

4. Quantitative measurement:

The services provided can be quantitatively measured.

The user is usually offered services based on certain charges.

Here, the application or resources are given as a utility on a pay-per-use basis. Thus, the use can be monitored and measured.

Not only the services are measureable, but also the link usage and several other parameters that support cloud applications can be measured.

This property of measuring the usage is usually not available in a web application and is a unique feature for cloud-based applications.

Cloud Application

5. ***On-demand service:*** *The cloud applications offer service to the user, on demand, that is, whenever the user requires it.*

The cloud service would allow the users to access web applications usually without any restrictions on time, duration, and type of device used.

Summary:

The previously mentioned properties are some of the features that make cloud a unique application platform.

These properties mentioned are specific to the cloud hence making it as one of the few technologies that allows application developers to suffice the user's needs seamlessly without any disruption.

Managing the Cloud

Cloud management is aimed at efficiently managing the cloud so as to maintain the QoS.

It is one of the prime jobs to be considered.

The whole cloud is dependent on the way it is managed.

Cloud management can be divided into two parts:

1. Managing the infrastructure of the cloud
2. Managing the cloud application

Managing the Cloud Infrastructure

The infrastructure of the cloud is considered to be the backbone of the cloud.

This component is mainly responsible for the QoS factor.

If the infrastructure is not properly managed, then the whole cloud can fail and QoS would be adversely affected.

The core of cloud management is resource management.

Resource management involves several internal tasks such as resource scheduling, provisioning, and load balancing.

These tasks are mainly managed by the cloud service provider's core software capabilities such as the cloud OS that is responsible for providing services to the cloud and that internally controls the cloud.

Managing the Cloud Infrastructure

A cloud infrastructure is a very complex system that consists of a lot of resources.

These resources are usually shared by several users.

Poor resource management may lead to several inefficiencies in terms of performance, functionality, and cost.

If a resource is not efficiently managed, the performance of the whole system is affected.

Performance is the most important aspect of the cloud, because everything in the cloud is dependent on the SLAs and the SLAs can be satisfied only if performance is good.

Managing the Cloud Infrastructure

Similarly, the basic functionality of the cloud should always be provided and considered at any cost.

Even if there is a small discrepancy in providing the functionality, the whole purpose of maintaining the cloud is futile.

A partially functional cloud would not satisfy the SLAs.

Lastly, the reason for which the cloud was developed was cost.

The cost is a very important criterion as far as the business prospects of the cloud are concerned.

On the part of the service providers, if they incur less cost for managing the cloud, then they would try to reduce the cost so as to get a strong user base

Managing the Cloud Infrastructure

Hence, a lot of users would use the services, improving their profit margin.

Similarly, if the cost of resource management is high, then definitely the cost of accessing the resources would be high and there is never a lossy business from any organization and so the service provider would not bear the cost and hence the users have to pay more.

Similarly, this would prove costly for service providers as they have a high chance of losing a wide user base, leading to only a marginal growth in the industry. And, competing with its industry rivals would become a big issue.

Hence, efficient management with less cost is required.

Managing the Cloud Infrastructure

At a higher level, other than these three issues, there are few more issues that depend on resource management.

These are power consumption and optimization of multiple objectives to further reduce the cost.

To accomplish these tasks, there are several approaches followed, namely, consolidation of server and storage workloads.

Consolidation would reduce the energy consumption and in some cases would increase the performance of the cloud.

server consolidation by definition is an approach to the efficient usage of computer server resources in order to reduce the total number of servers or server locations that an organization requires.

Managing the Cloud Infrastructure

The previously discussed prospects are mostly suitable for IaaS.

Similarly, there are different management methods that are followed for different types of service delivery models.

Each of the type has its own way of management. All the management methodologies are based on load fluctuation.

Load fluctuation is the point where the workload of the system changes continuously.

This is one of the important criteria and issues that should be considered for cloud applications.

Managing the Cloud Infrastructure

Load fluctuation can be divided into two types: predictable and unpredictable.

Predictable load fluctuations are easy to handle. The cloud can be preconfigured for handling such kind of fluctuations.

Whereas unpredictable load fluctuations are difficult to handle, ironically this is one of the reasons why cloud is preferred by several users.

Managing the Cloud Infrastructure

Cloud governance is another topic that is closely related to cloud management.

Cloud governance is different from cloud management.

Governance in general is a term in the corporate world that generally involves the process of creating value to an organization by creating strategic objectives that will lead to the growth of the company and would maintain a certain level of control over the company.

Similar to that, here cloud organization is involved.

Managing the Cloud Infrastructure

There are several aspects of cloud governance out of which SLAs are one of the important aspects.

SLAs are the set of rules that are defined between the user and cloud service provider that decide upon the QoS factor.

If SLAs are not followed, then the defaulter has to pay the penalty.

The whole cloud is governed by keeping these SLAs in mind.

Managing the Cloud Application

Business companies are increasingly looking to move or build their corporate applications on cloud platforms to improve agility or to meet dynamic requirements that exist in the globalization of businesses and responsiveness to market demands.

But, this shift or moving the applications to the cloud environment brings new complexities.

Applications become more composite and complex, which requires leveraging not only capabilities like storage and database offered by the cloud providers.

Managing the Cloud Application

But also third-party SaaS capabilities like e-mail and messaging.

So, understanding the availability of an application requires inspecting the infrastructure, the services it consumes, and the upkeep of the application.

The composite nature of cloud applications requires visibility into all the services to determine the overall availability and uptime.

Cloud application management is to address these issues and propose solutions to make it possible to have insight into the application that runs in the cloud, as well as implement or enforce enterprise policies like governance and auditing and environment management while the application is deployed in the cloud.

Managing the Cloud Application

These cloud-based monitoring and management services can collect a multitude of events, analyze them, and identify critical information that requires additional remedial actions like adjusting capacity or provisioning new services.

Additionally, application management has to be supported with tools and processes required for managing other environments that might coexist, enabling efficient operations.

Migrating Application to Cloud

Cloud migration encompasses moving one or more enterprise applications and their IT environments from the traditional hosting type to the cloud environment, either public, private, or hybrid.

Cloud migration presents an opportunity to significantly reduce costs incurred on applications.

This activity comprises, of different phases like evaluation, migration strategy, prototyping, provisioning, and testing.

Phases of Cloud Migration

Evaluation: Evaluation is carried out for all the components like current infrastructure and application architecture, environment in terms of compute, storage, monitoring, and management, SLAs, operational processes, financial considerations, risk, security, compliance, and licensing needs are identified to build a business case for moving to the cloud.

Migration strategy: Based on the evaluation, a migration strategy is drawn—a hotplug strategy is used where the applications and their data and interface dependencies are isolated and these applications can be operationalized all at once. A fusion strategy is used where the applications can be partially migrated; but for a portion of it, there are dependencies based on existing licenses, specialized server requirements like mainframes, or extensive interconnections with other applications.

Phases of Cloud Migration

Prototyping: Migration activity is preceded by a prototyping activity to validate and ensure that a small portion of the applications are tested on the cloud environment with test data setup.

Provisioning: Premigration optimizations identified are implemented. Cloud servers are provisioned for all the identified environments, necessary platform softwares and applications are deployed, configurations are tuned to match the new environment sizing, and databases and files are replicated.

All internal and external integration points are properly configured. Web services, batch jobs, and operation and management software are set up in the new environments

Phases of Cloud Migration

Testing: Postmigration tests are conducted to ensure that migration has been successful. Performance and load testing, failure and recovery testing, and scale-out testing are conducted against the expected traffic load and resource utilization levels.

Approaches for Cloud Migration

The following are the four broad approaches for cloud migration that have been adopted effectively by vendors:

1. **Migrate existing applications:** *Rebuild or rearchitect some or all the applications, taking advantage of some of the virtualization technologies around to accelerate the work. But, it requires top engineers to develop new functionality. This can be achieved over the course of several releases with the timing determined by customer demand.*
2. **Start from scratch:** *Rather than cannibalize sales, confuse customers with choice, and tie up engineers trying to rebuild existing application, it may be easier to start again. Many of the R&D decisions will be different now, and with some of the more sophisticated development environments, one can achieve more even with a small focused working team.*

Approaches for Cloud Migration

Separate company: One may want to create a whole new company with separate brand, management, R&D, and sales. The investment and internet protocol (IP) may come from the existing company, but many of the conflicts disappear once a new born in the cloud company is established. The separate company may even be a subsidiary of the existing company. What is important is that the new company can act, operate, and behave like a cloud-based start-up.

Buy an existing cloud vendor: For a large established vendor, buying a cloud-based competitor achieves two things. Firstly, it removes a competitor, and secondly, it enables the vendor to hit the ground running in the cloud space. The risk of course is that the innovation, drive, and operational approach of the cloud-based company are destroyed as it is merged into the larger acquirer.

**Thank You
For Your Attention...**